03 Person-Hours - Person-Hour Table Construction

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31 August 2026

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0.1 Introduction

This notebook expands each ICU stay into an hourly grid, attaches time-varying vitals and labs via LOCF (last-observation-carried-forward), and codes each row with its competing-event status (D'Agostino et al., 1990; Heyard et al., 2019).

Four mutually exclusive states at each hour:

Code	Event	Interpretation
0	In-progress	Patient remains at risk (censored at this hour)
1	Label onset	The variant's onset criteria are met in this hour
2	Discharge alive	ICU discharge without reaching onset
3	Death without onset	In-ICU mortality without prior onset

For variants A–C, state 1 is Sepsis-3 onset. For the alternative-label variants it is that variant's own onset: suspicion of infection (E) or acute organ dysfunction (D) under Protocol Amendment 2, and Sepsis-3 onset under the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ timing (F) under Protocol Amendment 4.

Treatment anchor timestamps (first antibiotic, first culture) are recorded for the treatment-anchored evaluation window (Kamran et al., 2024).

0.2 Load Cohort IDs

Cohort stay and admission IDs are collected so the large source tables can be filtered in a single DuckDB pass.

```
## Cohort: 65078 stays, 65078 hadm_ids
```

0.3 Load and Pre-Filter Large Tables

Chartevents and labevents are loaded via DuckDB with pushdown filters on item IDs and cohort membership. Temperature F is converted to C, and FiO_2 percentages are rescaled to fractions.

```
## chartevents filtered: 42,088,540 rows
```

```
## labevents filtered: 5,094,881 rows
```

0.4 Per-Hour Covariate Extraction

A dense hourly grid is built for each stay. Vitals and labs are aggregated to hourly means, then forward-filled (LOCF). Missingness indicators flag unmeasured hours. Six-hour rolling slopes capture vital-sign trends.

0.5 Treatment-Independent Deterioration Onset (Variant D)

Variant D (Protocol Amendment 2) has no suspicion anchor, so its onset cannot be assigned in `02_cohort_labels.Rmd`; it is derived here, where the hourly panel exists. Onset is the first hour after hour 1 at which the **physiology-only** SOFA score rises ≥ 2 points above the stay's admission baseline (hours 0–3, mirroring the admission baseline used by variants A and B).

"Physiology-only" means the vasopressor and catecholamine columns are withheld from `score_sofa()`, so the cardiovascular component is scored from MAP alone. The resulting label therefore contains no antibiotic order, no culture draw and no drug administration: nothing that records a clinician's decision to treat.

0.6 Standard Sepsis-3 Onset Timing (Variant F, Protocol Amendment 4)

Variants A–C take the onset to be the suspicion time t_{susp} whenever the SOFA criterion is met anywhere in the evaluation window. The standard operationalisation of Seymour et al. (2016), followed by the PhysioNet/CinC 2019 Challenge (Reyna et al., 2020), instead takes

$$t_{\text{onset}} = \min\left(t_{\text{susp}},\, t_{\text{SOFA}}\right),$$

where t_{SOFA} is the first hour inside the evaluation window $[t_{\text{susp}} - 48\,\text{h},\ t_{\text{susp}} + 24\,\text{h}]$ at which the SOFA score is already ≥ 2 points above the admission baseline. That branch is what allows an onset to precede the first antibiotic or culture; dropping it makes the pre-treatment window empty as a matter of definition rather than of fact.

Variant F restores it. Membership is **identical to Variant B** — the conjunction is unchanged, so exactly the same stays are septic — and only the onset timestamp moves, always weakly earlier. F therefore isolates the timing branch and nothing else.

Unlike Variant D, the SOFA score here is the full score, vasopressor components included, because that is what Seymour et al. specify. Vasopressor administration is itself a treatment, so an onset placed by the cardiovascular component is not treatment-free; it is only free of the *antibiotic-and-culture* anchor that defines the pre-treatment window. Variant D remains the fully treatment-independent comparison.

0.7 Build Person-Hour Tables Per Variant

For each label variant, the hourly panel is assembled, treatment anchors merged, competing events coded, and clinical scores (SOFA, NEWS2, qSOFA, SIRS) computed. Post-outcome rows are dropped. A temporal train/test split based on `anchor_year_group` follows Guo et al. (2022).

```
##     hr                500 values outside [20, 300] -> NA
##     resp_rate         128 values outside [0, 80] -> NA
##     spo2              731 values outside [50, 100] -> NA
##     map               2116 values outside [10, 250] -> NA
##     sbp               302 values outside [30, 300] -> NA
##     dbp_invasive      218 values outside [10, 200] -> NA
##     dbp_noninvasive   223 values outside [10, 200] -> NA
##     temp_c            830 values outside [25, 45] -> NA
##     fio2              1871 values outside [0.21, 1] -> NA
##     creatinine        21 values outside [0, 30] -> NA
##     bilirubin_total   7 values outside [0, 70] -> NA
##     platelets         3 values outside [0, 2000] -> NA
##     wbc               2 values outside [0, 500] -> NA
##     pf_ratio          124 values outside [10, 1000] -> NA
##     potassium         4 values outside [1, 12] -> NA
##   Plausibility filter (hourly panel): 7080 values set to NA
```

```
## Random split: 48808 train stays, 16270 test stays (seed=42, frac=0.25)
## Person-hour table VA: 2,853,777 rows, 72 cols
## Events: sepsis=5662, discharge=52892, death=5934, censored=2789289
##     hr                500 values outside [20, 300] -> NA
##     resp_rate         128 values outside [0, 80] -> NA
##     spo2              731 values outside [50, 100] -> NA
##     map               2116 values outside [10, 250] -> NA
##     sbp               302 values outside [30, 300] -> NA
##     dbp_invasive      218 values outside [10, 200] -> NA
##     dbp_noninvasive   223 values outside [10, 200] -> NA
##     temp_c            830 values outside [25, 45] -> NA
##     fio2              1871 values outside [0.21, 1] -> NA
##     creatinine        21 values outside [0, 30] -> NA
##     bilirubin_total   7 values outside [0, 70] -> NA
##     platelets         3 values outside [0, 2000] -> NA
##     wbc               2 values outside [0, 500] -> NA
##     pf_ratio          124 values outside [10, 1000] -> NA
##     potassium         4 values outside [1, 12] -> NA
##   Plausibility filter (hourly panel): 7080 values set to NA
```

```
## Random split: 48808 train stays, 16270 test stays (seed=42, frac=0.25)
## Person-hour table VB: 2,850,669 rows, 72 cols
## Events: sepsis=5887, discharge=52739, death=5892, censored=2786151
##     hr                500 values outside [20, 300] -> NA
##     resp_rate         128 values outside [0, 80] -> NA
##     spo2              731 values outside [50, 100] -> NA
##     map               2116 values outside [10, 250] -> NA
##     sbp               302 values outside [30, 300] -> NA
```

```
##     dbp_invasive     218 values outside [10, 200] -> NA
##     dbp_noninvasive  223 values outside [10, 200] -> NA
##     temp_c           830 values outside [25, 45] -> NA
##     fio2             1871 values outside [0.21, 1] -> NA
##     creatinine       21 values outside [0, 30] -> NA
##     bilirubin_total  7 values outside [0, 70] -> NA
##     platelets        3 values outside [0, 2000] -> NA
##     wbc              2 values outside [0, 500] -> NA
##     pf_ratio         124 values outside [10, 1000] -> NA
##     potassium        4 values outside [1, 12] -> NA
##   Plausibility filter (hourly panel): 7080 values set to NA

## Random split: 48808 train stays, 16270 test stays (seed=42, frac=0.25)
## Person-hour table VC: 2,625,204 rows, 72 cols
## Events: sepsis=11531, discharge=47541, death=4956, censored=2561176
##     hr               500 values outside [20, 300] -> NA
##     resp_rate        128 values outside [0, 80] -> NA
##     spo2             731 values outside [50, 100] -> NA
##     map              2116 values outside [10, 250] -> NA
##     sbp              302 values outside [30, 300] -> NA
##     dbp_invasive     218 values outside [10, 200] -> NA
##     dbp_noninvasive  223 values outside [10, 200] -> NA
##     temp_c           830 values outside [25, 45] -> NA
##     fio2             1871 values outside [0.21, 1] -> NA
##     creatinine       21 values outside [0, 30] -> NA
##     bilirubin_total  7 values outside [0, 70] -> NA
##     platelets        3 values outside [0, 2000] -> NA
##     wbc              2 values outside [0, 500] -> NA
##     pf_ratio         124 values outside [10, 1000] -> NA
##     potassium        4 values outside [1, 12] -> NA
##   Plausibility filter (hourly panel): 7080 values set to NA

##   Variant D: 27,209 / 65,078 stays deteriorate (41.8%), median onset hour 9.0
## Random split: 48808 train stays, 16270 test stays (seed=42, frac=0.25)
## Person-hour table VD: 1,998,748 rows, 72 cols
## Events: sepsis=27209, discharge=35248, death=2621, censored=1933670
##     hr               500 values outside [20, 300] -> NA
##     resp_rate        128 values outside [0, 80] -> NA
##     spo2             731 values outside [50, 100] -> NA
##     map              2116 values outside [10, 250] -> NA
##     sbp              302 values outside [30, 300] -> NA
##     dbp_invasive     218 values outside [10, 200] -> NA
##     dbp_noninvasive  223 values outside [10, 200] -> NA
##     temp_c           830 values outside [25, 45] -> NA
##     fio2             1871 values outside [0.21, 1] -> NA
##     creatinine       21 values outside [0, 30] -> NA
##     bilirubin_total  7 values outside [0, 70] -> NA
##     platelets        3 values outside [0, 2000] -> NA
##     wbc              2 values outside [0, 500] -> NA
##     pf_ratio         124 values outside [10, 1000] -> NA
```

```
##     potassium        4 values outside [1, 12] -> NA
##   Plausibility filter (hourly panel): 7080 values set to NA
## Random split: 48808 train stays, 16270 test stays (seed=42, frac=0.25)
## Person-hour table VE: 2,392,855 rows, 72 cols
## Events: sepsis=17251, discharge=42412, death=3784, censored=2329408
##     hr               500 values outside [20, 300] -> NA
##     resp_rate        128 values outside [0, 80] -> NA
##     spo2             731 values outside [50, 100] -> NA
##     map              2116 values outside [10, 250] -> NA
##     sbp              302 values outside [30, 300] -> NA
##     dbp_invasive     218 values outside [10, 200] -> NA
##     dbp_noninvasive  223 values outside [10, 200] -> NA
##     temp_c           830 values outside [25, 45] -> NA
##     fio2             1871 values outside [0.21, 1] -> NA
##     creatinine       21 values outside [0, 30] -> NA
##     bilirubin_total  7 values outside [0, 70] -> NA
##     platelets        3 values outside [0, 2000] -> NA
##     wbc              2 values outside [0, 500] -> NA
##     pf_ratio         124 values outside [10, 1000] -> NA
##     potassium        4 values outside [1, 12] -> NA
##   Plausibility filter (hourly panel): 7080 values set to NA
##   Variant F: 6,447 onsets (membership identical to B); 1,738 moved earlier (27.0%), median sh
## Random split: 48808 train stays, 16270 test stays (seed=42, frac=0.25)
## Variant F: 419 of 6447 onsets fall before the treatment anchor (was 0 under conjunction timin
## Person-hour table VF: 2,819,734 rows, 74 cols
## Events: sepsis=6074, discharge=52739, death=5892, censored=2755029
```